MUSIC × LEARNING × TECHNOLOGY

Your syllabus. Your *soundtrack.*

Meet **Syllabeat**—an Android learning experience that turns academic concepts into songs, then turns listening into practice.

130
curriculum topics

9
offline recordings

5
learning steps

A five-member NMIMS student project

Presented by Dhvani Tandel · MBATech CE

OPEN THE ONLINE PRESENTATION

dhvicore.github.io/designexp/dhvani



BUILT FOR THE WAY STUDENTS LISTEN

01 / THE IDEA

Make revision *worth replaying.*

Formulas and definitions can be difficult to revisit. Our idea combines a memorable song with an activity that checks understanding.

S

Song



Original educational lyrics turn the key concept into a verse and chorus.

LISTEN & REPEAT

A

Activity

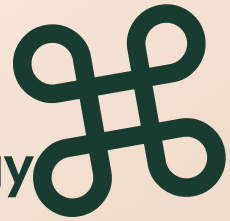


Blanks, multiple-choice questions and line ordering make the student participate.

RECALL & CHECK

T

Technology



One Android app connects topic selection, lyrics, audio and feedback.

ONE LEARNING JOURNEY

OUR HYPOTHESIS

Music can make revision more engaging. Whether it improves long-term retention still needs a student study.

SEMESTER / JUNE–NOVEMBER 2026

The semester *at a glance*

Research and documentation became a curriculum dataset, then a working Android learning app.

JUN

Research

Investigated music-based learning, AI tools and the supporting research files.

JUL

Dataset

Documented Calculus topics, formulas, applications and music categories.

AUG

Mapping & AI

Linked topics to song references and investigated lyric-generation tools.

SEP

Preparation

Continued research and prepared the project's content and documentation files.

OCT

Implementation

Developed the prototype and Android app, audio playback and activities.

NOV

Integration

Refined Syllabeat, added Economics and updated the report and presentation.

LOGBOOK REFERENCE

20 Jul · 29 Jul · 11 Aug · 18 Aug. June and September summaries follow Dhvani's account; no extra meeting dates are inferred.

SEMESTER / RESEARCH & DATA

Research first
Then structured content

June research and file preparation supported the detailed dataset work recorded in July.

JUNE / RESEARCH FILES

Understanding the approach

- Studied music generation and educational uses of language models.
- Organised papers, reference notes and project documentation.
- Used the Song–Activity–Technology framework to guide the learning experience.

JULY / DATED LOGBOOK

Building the Calculus data

20 July · Week 4

Syllabus specification: concepts, exact formulas and engineering applications.

29 July · Week 5

English/Hindi song types and suitable music categories.

Three linked artifacts: syllabus specification, song/topic mapping and per-topic generation briefs.

Sources: supplied report §2 and §3; Calculus_Dataset_Findings.md; June research confirmed by Dhvani.

SEMESTER / CONTENT & AI RESEARCH

A formula needs *the right words*

AUGUST / LOGBOOK WEEKS 6 & 7

11 August · Week 6

Matched the Calculus topics with music styles and English/Hindi references.

18 August · Week 7

Reviewed GPT, Llama and Claude for educational lyrics, plus Suno, MagicSchool and MusicGen.

Next task recorded: prepare Unit 1 lyrics grounded in the syllabus and make sample songs.

Topic mapping established the link between academic content, language and musical style.

SEPTEMBER / CONTINUED PREPARATION

Files ready for implementation

Continued research and preparation of the project files, including lyric material, findings, tool guidance and documentation.

Core rule: keep the theorem's assumptions and formula intact when adapting words to music.

Content accuracy comes before rhyme

Source: Logbook_Week7_Entry.md, including prior meeting dates; September summary confirmed by Dhvani. File completion dates are not assigned individually.

RESEARCH / WHY THIS APPROACH

Music is the hook. *Understanding is the goal.*

The original literature review informed the SAT idea. It does not prove that our app improves marks or long-term retention.

LEARNING DESIGN

Pair songs with action

Use a chorus for recall, then ask students to retrieve facts through blanks, choices and sequencing.

Active-learning literature informs the approach; the app still needs evaluation.

CONTENT ACCURACY

Ground every lyric

Start from the syllabus explanation, exact formula and required conditions. Review translations and singing pronunciation.

Fluent lyrics can still contain academic errors.

TOOL EXPLORATION

Choose what works

Suno supplied the final saved recordings. Colab / ACE-Step was experimental. MusicGen and MagicSchool were researched.

No custom singing model was successfully trained for the final demo.

WHAT WE WILL MEASURE

Compare song + activity with conventional revision, using topic quizzes immediately and again after a delay.

PROTOTYPE / PYTHON & STREAMLIT

Testing the idea *before Android*

Calculus Song Studio explored the full SAT experience and revealed the limitations of synthetic singing.

WHAT WE EXPLORED

Song and activity together

- Unit/topic/language selection and educational lyrics.
- Speech voices shaped onto notes with Praat/PSOLA.
- Blanks, formula choices and ordering activities.
- Whisper sing-back checking and a Photosynthesis demonstration.

WHAT CHANGED FOR ANDROID

Clarity became the priority

Robotic voices and weak word clarity led to saved Suno recordings for the Calculus demo.

Colab/ACE-Step provided another experimental route, but exact tune control and service reliability remained unresolved.

Whisper sing-back and the Biology demo remain prototype explorations, not shipped Android features.

References: original report §4, prototype/README.md and the current Android implementation. No custom singing model was trained for the final demo.

02 / CURRICULUM FIRST

Real syllabus. *Structured for songs.*

The work began with syllabus collection, unit-wise topic segregation, key terms and song-reference mapping—not audio alone.

CALCULUS

38 topics · 5 units

Differential calculus, partial differentiation, integrals, multiple integrals and vector calculus.



Topic count by unit, U1 → U5

CHEMISTRY

60 topics · 6 units + supplements

Atomic structure, gas laws, water, materials, electrochemistry, acids and bases.

English | हिन्दी | मराठी

THE SHARED DATA CONTRACT

01 Curriculum content
Subject → unit → topic → explanation + formula / keywords

↓

02 Music mapping
Category → language → reference song → musical style

↓

03 Learning output
Educational lyrics → recording → timed cues → practice

Stable topic IDs keep songs, lyric versions and quiz content connected. Bundled JSON makes the demo available offline.

ECONOMICS / MANAGEMENT 32 topics · 7 units · English, Hindi and Marathi · added November

One topic.
Five simple steps.

Sign up or log in, then follow a guided learning flow.

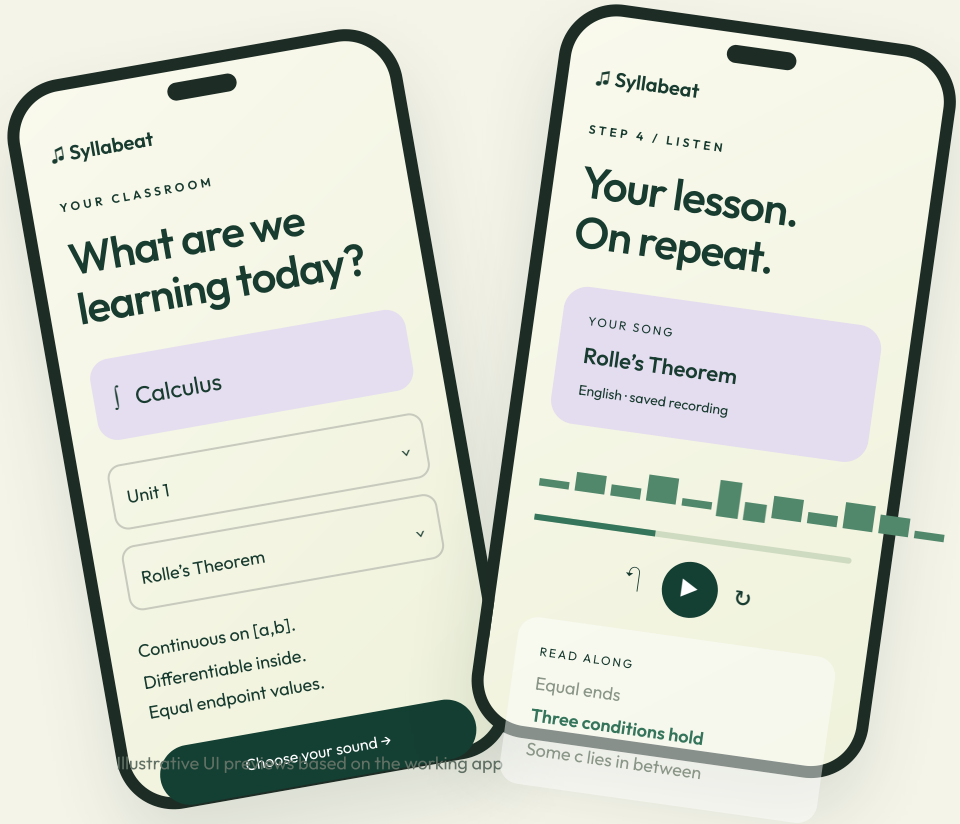
01 Learn
Select subject, unit and topic; review the concept.

02 Choose your sound
Calculus: Hindi, English, bhajan or lo-fi. Chemistry and Economics: English, Hindi or Marathi.

03 Review the lyrics
Prepare a curriculum-based draft or load the saved recording's lyrics.

04 Listen & follow
Play offline, pause, seek, replay and read highlighted lines.

05 Practise
Answer three activity types, get a score and try again.



04 / PRESS PLAY

Same theorem.
A different rhythm.



Original educational lyrics

Saved Suno-generated recordings

Try the saved Rolle's Theorem recordings. Each category has its own audio file and line timings.

English | Hindi | Bhajan | Lo-fi

LIVE AUDIO SAMPLE

English · Rolle's Theorem

No generation server needed for these saved songs.

FOLLOW THE LYRICS

Press play to hear the selected recording.

Automatic line timing is approximate. Song names in the app are references; exact matching to a commercial melody is not verified.

4 Rolle's Theorem versions **4** Beta Function versions **1** Dalton's Atomic Theory English version

NOVEMBER / LATEST INTEGRATION

Economics joins *the same learning flow*

Krishhma Vira's Economics/Management contribution adds 32 topics across seven units, with English, Hindi and Marathi content.

CONTENT IN THE APP

Seven units

Introduction, markets/macroeconomics, central banking, production, management, organising and controlling.

Each topic connects its explanation and key terms to language-specific song suggestions, editable lyrics and practice.

Twelve topics include supplied English lyrics. Other drafts use templates based on the contributed content.

ON-DEVICE AUDIO

A second audio route

1. Android text-to-speech records each lyric line.
2. Pitch and syllable processing arrange the voice over a generated beat.
3. The app saves a WAV and line timings for playback and read-along lyrics.

No Suno or Colab service is required for this route. Voice quality depends on Android's installed voices.

STYLE REFERENCES

Named songs guide style and tempo. This engine composes an original melody, rather than reproducing the selected commercial recording.

PRACTICE / THE A IN SAT

After the chorus, *check the concept.*

ROLLE'S THEOREM / EXAMPLE

A song is a starting point

The function must be **continuous** on the closed interval [a,b].

It must be differentiable inside, and f(a) = f(b).

$$\exists\, c \in (a,b),\ f'(c) = 0$$

All three assumptions are required before applying the conclusion.

Step 5 turns the listening session into three scored tasks. Students receive feedback and can retry.

01 **Fill in the blanks**

Recall important words from the lesson instead of only recognising them.

02 **Choose the right answer**

Pick the correct formula or relevant key terms for the selected subject.

03 **Put the lines in order**

Rebuild the lesson, check the feedback and retry for a better score.

CURRENT SCOPE

Activities run offline and are currently in English. Sing-back assessment and scheduled revision remain future Android features.

05 / UNDER THE HOOD

A simple interface. *A connected system.*

Built in Android Studio with Kotlin and Jetpack Compose; tested on an Android emulator.

ANDROID / KOTLIN

Jetpack Compose interface

Login · topic selection · music choices · lyrics · player · activities

Curriculum

Bundled JSON records

Stable topic IDs

Multilingual content

Accounts

Device-local SQLite

Salted PBKDF2 hashes

Remembered email

Audio + lyrics

Original MP3 assets

Buffered Media3 playback

Per-recording line cues

Practice

Content-based blanks

Formula / keyword quiz

Line order + feedback

DESKTOP AUTHORING

Reviewed syllabus lyrics → Suno → downloaded MP3 → local Whisper timing → Android assets

No transcription or model inference runs during saved-song playback.

ECONOMICS ROUTE: topic + lyrics → Android TTS → pitch/beat renderer → WAV + line cues → Media3. Installed voices are preferred; a network voice may require internet.

TECHNICAL / SHORT CODE WALKTHROUGH

What the main *files actually do*

The app separates the screens, curriculum, accounts, song creation and playback so they can develop independently.

`SongStudio.kt`

Controls the five learning steps and remembers the selected subject, topic, category and lyrics.

`Curriculum.kt + JSON assets`

Loads syllabus records and stable topic IDs. Chemistry and Economics adapters map team data into one common structure.

`AccountStore.kt`

Stores device-local accounts with salted password hashes. Remembers the email while requiring the password again.

`EconomicsAudio.kt`

Records TTS lines, runs the imported pitch/beat renderer and saves a WAV with lyric cues.

`LocalSongEngine.kt + TimedLyrics.kt`

Uses Media3 to play local audio. The playback clock chooses which lyric line to highlight.

`LearningActivity.kt`

Creates blanks, formula/key-term choices and ordering tasks, then checks answers and offers a retry.

06 / WHAT WE BUILT & LEARNED

From an experiment *to a working app.*

The original report and deck documented the Calculus prototype. The current implementation adds a complete Android learning journey.

01 / FOUNDATION

Data & research

Collected syllabus topics, formulas, key terms and music references. Developed the Song–Activity–Technology framework.

02 / PROTOTYPE

Python + Streamlit

Explored text-to-speech, pitch shaping, melodies and quizzes. Robotic vocals exposed the gap between speech and natural singing.

03 / EXPLORATION

AI audio trials

Tested a GPU/Colab music workflow. Singing quality, service connectivity and exact melody control remained challenging.

04 / WORKING DEMO

Android + saved songs

Integrated saved songs, Chemistry, lyric cues and quizzes in October. Added Economics and on-device generation in November.

RELIABILITY

Buffered playback and in-place updates preserve the demo and local accounts.

QUALITY CHECKS

Build/lint, topic mappings, audio continuity, seeking and lyric timing checks.

HONEST LIMIT

Calculus/Chemistry use saved recordings. Economics generates TTS-based songs on the device; natural singing quality remains an open issue.

VALIDATION / WHAT WORKS TODAY

Test the experience. *Be clear about the limits.*

We checked the implementation, while keeping technical playback separate from musical quality and learning effectiveness.

IMPLEMENTED & CHECKED

A working Android demo

- Build and lint checks
- Curriculum and song-selection mappings
- Account persistence across normal updates
- Original-file playback, pause, seek and replay
- Line-cue order and recording boundaries
- Quiz content, answer handling and feedback

OPEN LIMITATIONS

A prototype, not a finished service

- 9 bundled recordings plus Economics generation
- Automatic lyric timing is approximate
- One short recording cuts off mid-refrain
- No verified exact commercial-song covers
- No SAP login, cloud sync or full library yet
- No measured retention benefit yet

PLAYBACK TROUBLESHOOTING

We increased buffering and switched the Mac emulator from software to hardware graphics. Emulator performance still depends on the host device.

FUTURE / THE NEXT CHAPTER

A soundtrack for *every subject.*

Proposed next steps build on the same curriculum → lyrics → audio → activity structure.

01

EXPAND THE FOUNDATION

More subjects. More students.

Add Physics, Biology and further syllabus data. Economics is now integrated. Introduce verified branch → year → semester mappings.

- Physics
- Biology
- More curricula
- All subjects

02

PERSONALISE THE EXPERIENCE

Smarter practice. Better continuity.

Add song history, cloud sync, teacher content review, more languages and spaced-repetition reminders.

- Saved library
- Progress
- Regional languages

03

IMPROVE & EVALUATE

Better singing. Measured learning.

Connect a supported singing API, explore licensed melody conditioning and sing-back feedback, then test recall with students.

- Audio quality
- Sing-back
- Retention study

PLANNED, NOT YET SHIPPED

SAP/college portal login needs institutional access. Exact commercial-song melody control needs suitable rights, references and further validation.

SYLLABEAT / THE NEXT WAY TO REVISE

Learn the concept. *Feel the rhythm.* Check what stays.

A working Android prototype bringing curriculum, educational music, read-along lyrics and active practice into one experience.

3 integrated subjects

130 curriculum topics

9 saved recordings

NMIMS MPSTME · Design Experience

Faculty mentors: Dr. Jaychand Upadhyay & Dr. Dhirendra Mishra